

# Mohamed Shereef CH

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## EDUCATION

- Bachelor of Technology in Information Technology** *Expected May 2026*  
MEA Engineering College, Kerala

## TECHNICAL SKILLS

- Languages:** Python, JavaScript, TypeScript, Java (fundamentals), C (fundamentals)
- Core CS:** Data Structures & Algorithms, Object-Oriented Design, System Design, Operating Systems, Computer Networks
- Security:** JWT Authentication, Row-Level Security (RLS), CSP Headers, CORS Policies, Rate Limiting, Server-Side Token Validation, Cryptographic Code Generation
- Backend:** FastAPI, Node.js, REST API Design, Asynchronous Processing, Supabase Edge Functions, PostgreSQL
- Frontend:** React, TypeScript, Tailwind CSS, Vite, Zustand
- Cloud & DevOps:** Docker, Git, GitHub Actions (CI/CD), Vercel, Serverless Architecture, WebRTC, WebSocket
- AI/ML & Tools:** GitHub Copilot, LLM APIs (Gemini, OpenRouter), PyTorch, EfficientNet, Grad-CAM

## PROJECTS

- Kallanum Policeum — Serverless Multiplayer Game with Security-First Architecture** *React, TypeScript, Supabase, Supabase Edge Functions, Ably, Agora, Vercel | GitHub*
  - Browser-based multiplayer game (3–15 players) running entirely on serverless infrastructure; no dedicated backend server; all game logic executes in Supabase Edge Functions
  - Applied layered security across the stack: JWT validation on every Edge Function call, RLS policies on all database tables, server-side role assignment with no client trust, rate limiting, CSP headers, and CORS restrictions
  - Implemented realtime game state sync via Supabase Realtime, pub/sub messaging via Ably, and WebRTC voice chat via Agora with server-generated tokens to prevent client-side token abuse
  - Built room management with cryptographically generated codes, a dynamic role assignment engine, and a complete Web Audio API sound system
- Book2Vision — Async Backend Pipeline for AI Multimedia Generation** *Python, FastAPI, Gemini API, ElevenLabs, REST APIs | GitHub*
  - Built a multi-stage Python backend that takes book files (PDF, EPUB, TXT) through OCR, text chunking, LLM summarization, TTS audio, and image generation in sequence
  - Used FastAPI with async task handling to manage long-running, high-latency jobs without blocking; supports concurrent requests across multiple content formats
  - Integrated ElevenLabs, Gemini, and image generation APIs through a unified service layer designed for easy extension or swapping of individual providers
  - Structured the codebase into clearly separated modules following OOP principles; each pipeline stage is independently testable and replaceable
- CPMS — Campus Placement Management System** *JavaScript, HTML5, CSS3, PostgreSQL, Supabase | GitHub*
  - Multi-role platform with separate workflows for students, recruiters, and admins; built end-to-end from schema design to frontend
  - Designed a normalized PostgreSQL schema with row-level security; each role accesses only the records its policy permits, enforced at the database layer
  - Backend handles session management, PDF parsing, and structured data operations through RESTful endpoints with consistent error handling and input validation
  - Integrated Gemini API for automated resume scoring with OpenRouter as a fallback; the service layer abstracts provider details from the rest of the application
- ACLInsight-Net — Deep Learning MRI Diagnostic System** *Python, PyTorch, EfficientNet-B4, Grad-CAM | GitHub*
  - Multi-plane MRI fusion model (axial, coronal, sagittal) for ACL tear detection using EfficientNet-B4; validated with 5-fold cross-validation, achieving AUC  $\approx 0.90$
  - Refactored the full training pipeline into 8 modular Python files with clear boundaries between data loading, model definition, training loop, and evaluation
  - Applied Grad-CAM to visualize which regions drove model decisions—useful for catching spurious correlations during development
  - Debugged CUDA out-of-memory crashes, RAM overflow during batch loading, and checkpoint resumption across interrupted training runs; paper in preparation for IEEE submission

## RELEVANT COURSEWORK

Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, Software Engineering, Cryptography & Network Security, Machine Learning Fundamentals, System Design

## ADDITIONAL INFORMATION

- Languages:** English (professional), Malayalam (native), Hindi
- Interests:** Cloud security, distributed systems, serverless architecture, backend engineering